W-C-582A
June 25, 1981
SUPERSEDING
Fed. Spec. W-C-582
July 7, 1964

FEDERAL SPECIFICATION

CONDUIT, RACEWAY, METAL, AND FITTINGS; SURFACE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specificatiaon covers metal raceway conduits (hereinafter referred to as raceways) and fittings for surface wiring as distinguished from rigid metal conduit and electrical metallic tubing. It does not cover devices for use in hazardous locations, as defined in the National Electrical Code.

1.2 Classification.

1.2.1 Types. Raceways and fittings shall be of the following types, as specified (see 6.1):

Type I - Wall or ceiling installations.
Type II - Floor installation.
Type III - Wall or ceiling installation with integral receptacles. Switches, circuit breakers and ground fault circuit interrupters available (see 6.1).

1.2.2 Sizes.

1.2.2.1 Type I and II raceways. Type I and II raceways and fittings shall be furnished, as specified (see 6.1), in the sizes described. Each size shall accommodate not less than the number and American Wire Gage (AWG) sizes of rubber insulated (R) or thermoplastic insulated (T) conductors indicated.

1.2.2.2 Type III raceways. Type III raceways shall be furnished in 36- or 72-inch lengths as specified (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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Federal Specifications:

W-C-375 - Circuit Breakers, Molded Case, Branch Circuit and Service (General Specification).
W-C-596 - Connector, Electrical, Power, General Specification for.
W-S-896 - Switches, Toggle (Toggle and Lock), Flush Mounted (General Specification).

Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Underwriters' Laboratories, Inc., (UL) Standards:

UL 5 - Surface Metal Raceways and Fittings.
UL 498 - Attachment Plugs and Receptacles.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 UL label or listing mark. Each contractor shall submit to the contracting agency proof that the raceways and fittings he proposes to supply under this specification conform to the requirements of UL 5 and applicable bulletins. The UL label or listing mark may be accepted as evidence that the raceways and fittings conform to UL requirements.

3.1.2 Alternate certification. In lieu of the label or listing mark, the contractor shall submit independent proof, satisfactory to the contracting agency, that the raceways and fittings conform to the applicable requirements of the published standards including methods of tests of UL 5.

3.1.3 Additional requirements. In addition to compliance with 3.1.1 and 3.1.2, the raceways and fittings shall comply with the additional requirements specified herein.

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3 Design and construction. Raceways, fittings, and raceway receptacles shall be of the design, construction, and physical dimensions as specified herein (see 6.1).

3.3.1 Raceways. Raceways shall be restricted to 50/60 Hertz (Hz) use. Design shall conform to UL 5.

3.3.1.1 Type I. The type I raceway shall be oblong in cross section, have a blank cover, and shall be used in installations where the raceways are mounted on the ceiling or along walls.

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3.3.1.2 Type II. The type II raceway shall be flat and narrow in cross section, have either a snap-on or screw-on cover, and shall be used for over-floor mounting.

3.3.1.3 Type III. The type III raceway shall be oblong in cross section, have integral receptacles mounted on a snap-on cover, and shall be used on ceiling or wall installations. The type III raceway shall be equipped with the number of grounded receptacles as specified (see 6.1). Grounding shall be accomplished by connection to a bare ground wire run in the raceway. The 72-inch length shall have not more than 12 receptacles nor less than four receptacles evenly spaced. The 36-inch length shall have not more than six receptacles. Associated wiring shall not be smaller than 12 AWG.

3.3.2 Fittings. Fittings shall comply with the design requirements of the applicable raceway and shall conform to UL 5.

3.3.3 Receptacles. Receptacles used shall be of the 50/60 Hz grounding type and shall conform to W-C-596. Such receptacles not covered by W-C-596 shall conform to UL 498.

3.3.3.1 Color. Unless otherwise specified, the color of the receptacle shall be black (see 6.1).

3.3.3.2 Rating (type III). Unless otherwise specified, receptacles used shall be rated either at 15 amperes, 125 volts or 15 amperes, 250 volts (see 6.1).

3.3.3.3 Mating plugs. Mating plugs shall be of the identical rating as the applicable receptacles, and shall conform to W-C-596. Those plugs not covered by W-C-596 shall conform to UL 498.

3.3.4 Circuit breakers. Circuit breakers shall be in accordance with W-C-375.

3.3.5 Switches. Switches shall be in accordance with W-S-986.

3.4 Performance. Raceways, fittings, and raceway receptacles shall meet the performance requirements specified herein.

3.4.1 Raceways and fittings. When tested in accordance with 4.6, the raceways and fittings shall meet the following requirements as specified in UL 5.

3.4.2 Receptacles. Receptacles shall meet the performance requirements of W-C-596. Receptacles not covered in W-C-596 shall meet the performance requirements of UL 498.

3.5 Workmanship.

3.5.1 Metal fabrication. Metal used in the fabrication of raceways and fittings shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of

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metal, may be employed on steel instead of shearing or sawing. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.5.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided, when necessary, to reduce stress concentration. Welding shall be in accordance with a nationally recognized welding code.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. An inspection lot shall consist of all raceways and fittings of the same type offered for delivery at the same.

4.3 Sampling. A random sample of raceways or fittings shall be selected from each lot in accordance with MIL-STD-105.

4.4 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5 Examination. Devices selected in accordance with 4.3 shall be examined to verify that the materials design, construction, physical dimensions, and workmanship are in accordance with the applicable requirements of section 3. The inspection level shall be II with an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.6 Tests. Raceways and fittings shall be tested as specified in UL 5 in the following sequence:

(a) Fixture support test.
(b) Chronic acid test.
(c) Electrical resistance.

4.7 Preparation for delivery inspection. An examination shall be made to determine compliance with the requirements of section 5. A sample unit shall

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be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Items shall be cleaned and dried by an applicable process in MIL-P-116.

5.1.1.2 Unit packing. Items shall be unit packed method III, in accordance with MIL-P-116. The unit container shall be a box selected from table IV of MIL-STD-794.

5.1.2 Commercial. Items shall be packaged in accordance with the contractor's standard practice in a manner to prevent deterioration and damage during shipment, handling, and storage.

5.2 Packing. Packing shall be levels A, B, or commercial, as specified (see 6.1).

5.2.1 Levels A and B. Items shall be packed in accordance with the packing requirements of MIL-STD-794. Containers shall be selected from table II of MIL-STD-794 for the appropriate level of packing.

5.2.2 Commercial. Items shall be packed in a manner which will insure arrival at destination in a satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size, and for type III, unit length (see 1.2.1 and 1.2.2.1).
(c) Number of receptacles (see 3.3).
(d) Whether receptacles shall be other than black (see 3.3.3.1).
(e) When rating is other than specified in 3.3.3.2.
(f) Levels of packaging and packing required (see 5.1 and 5.2).

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